S3 Supplementary QC and Robustness

Support boundaries and L4 certificate audit

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S3.1 Coverage

Chart-Jacobian alpha is finite on 134/134 recordings. Lag-1 one-step quantities are computed on 121/134 recordings; history-conditioned L2 is computed on 115/134. Lag-2 is absent. These are support facts, not successful imputation targets.

S3.2 L4 certificate failures

Subject	Task	Gate	Failure
`sub-S200121`	Rest	F	`post_dtype_psd_failure`
`sub-S200214`	GG	F	`post_dtype_psd_failure`
`sub-S210203`	CC	F	`post_dtype_psd_failure`

Table 1: All missing finite `reach_a_q` certificates. One record falls in each task and each subject is distinct.

S3.3 Degenerate readout

Of 131 finite `reach_d_eff` values, 109 (83.2%) lie within $10^{-}6$ of 1; the median and minimum are 1 and the maximum is 1.31377. This rank-degenerate ellipsoid summary cannot support a capacity-dimension contrast. Apparent Wilcoxon significance on residual numerical variation is therefore method-limited and excluded from the article's substantive claims.

S3.4 Next QC gate

The bounded audit is complete. The 13 recordings comprise 12 Rest and 1 CC (0 GG), all on a regular 4-s grid with 120–260 lag-1 pairs. Each fails the existing out-of-fold fidelity gate (`median(rel-MSE fold 1, fold 2) < 0.9`) with `one_step_fit_not_better_than_baseline`; reproduced fold scores agree within 1.338e-07. Diagnostic threshold sweeps (not production changes) identify 126/134 at 0.95, 128/134 at 1.0, and 131/134 at 1.5. This identifies a reproducible support boundary, not a reason to fill absent lags or reopen the estimator.

QC verdict

PARTIAL PASS for coverage reporting and L4 certificate transparency. Lag-1 support remains partial; lag-2 is unavailable; `reach_d_eff` is method-limited. No claim of a complete one-step or dimension analysis is licensed.

Bibliography